

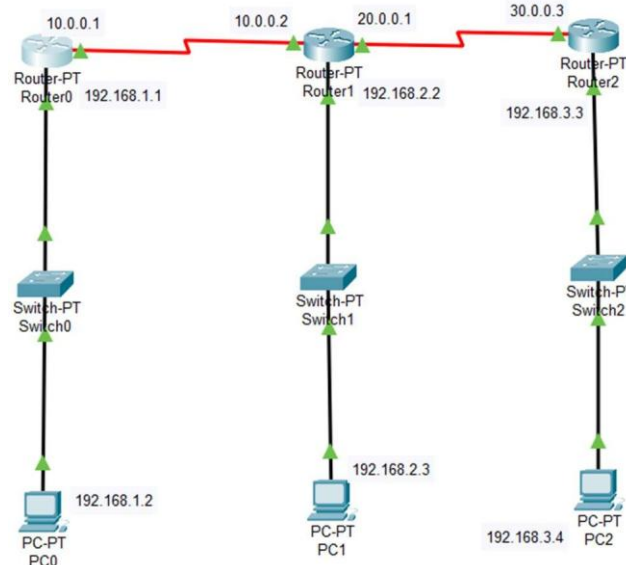
EXPERIMENT NO:-06

AIM:- To study about the configuration of distance vector RIP routing protocol by using Cisco Packet Tracer.

● STEPS :

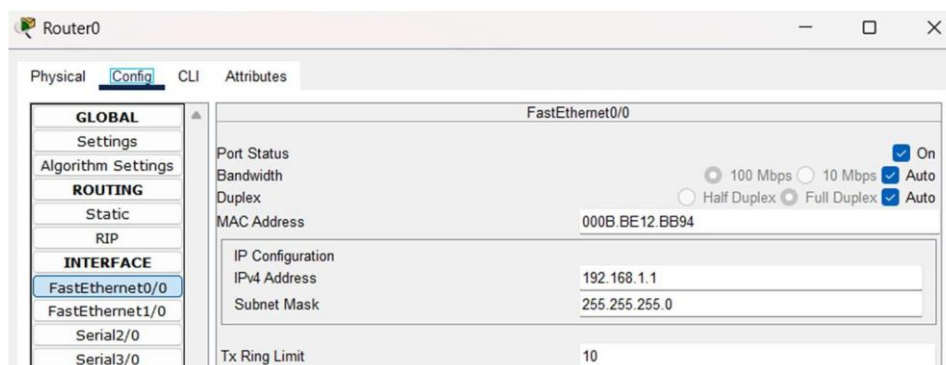
1. Setup the Network:

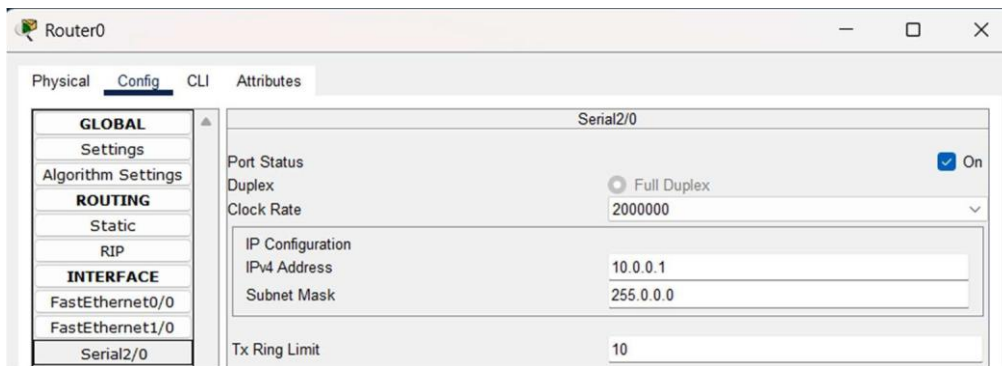
- Create a connection similar to the one shown in the reference diagram.
- Arrange routers, switches, and PCs, and connect them using appropriate cables.



2. Assign IP Addresses:

- Assign IP addresses to all routers and PCs step by step.
- For each PC, go to the **Desktop Tab > IP Configuration** and configure the IPv4 Address, Subnet Mask, and Default Gateway.





3. RIP Routing Configuration:

- Configure RIP on all routers.
- Use the following commands for each router:

- On **Router0**:

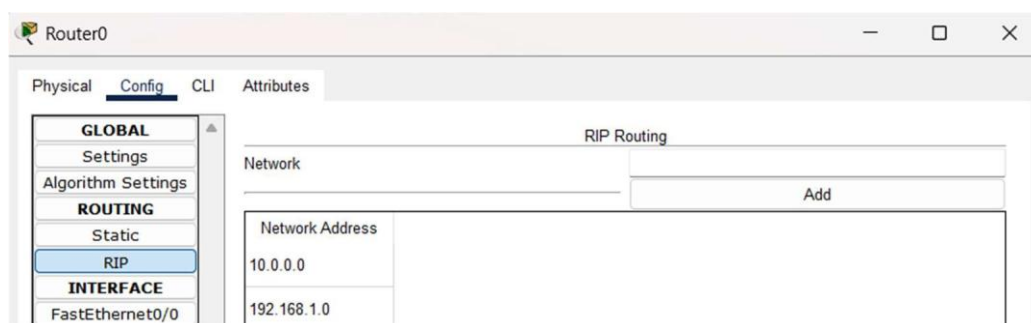
```
router rip
version 2
network 192.168.1.0
network 10.0.0.0
```

- On **Router1**:

```
router rip
version 2
network 10.0.0.0
network 20.0.0.0
network 192.168.2.0
```

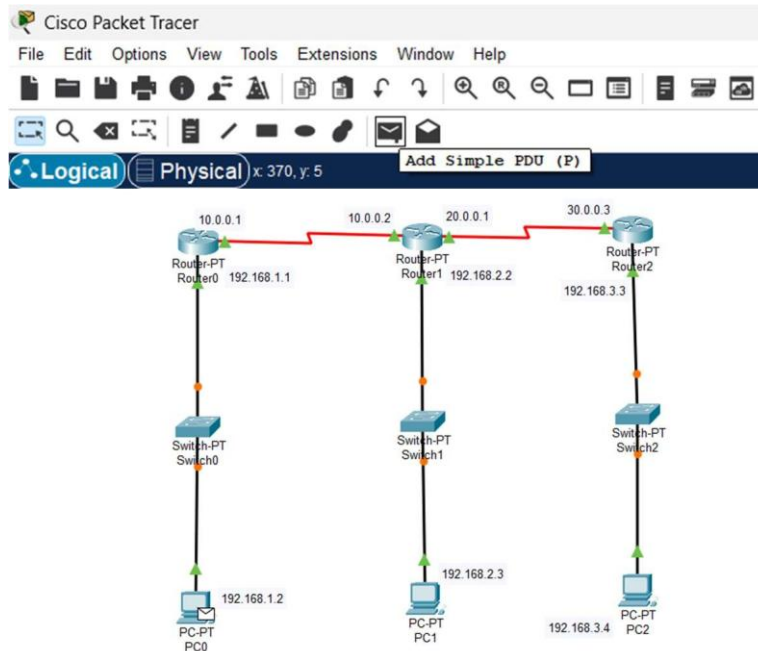
- On **Router2**:

```
router rip
version 2
network 20.0.0.0
network 192.168.3.0
```



4. Send Data:

- Use the **Add Simple PDU** tool.
- Select a PC and send a message to a PC in a network connected to a different router (e.g., from PC0 to PC2).



5. Simulation Mode:

- Switch to **Simulation Mode**.
- Run the simulation to observe the RIP routing process and ensure successful packet delivery.
- Event List

Reset Simulation ☒ Constant Delay Captured to: (no captures)

Play Controls

Event List Filters - Visible Events

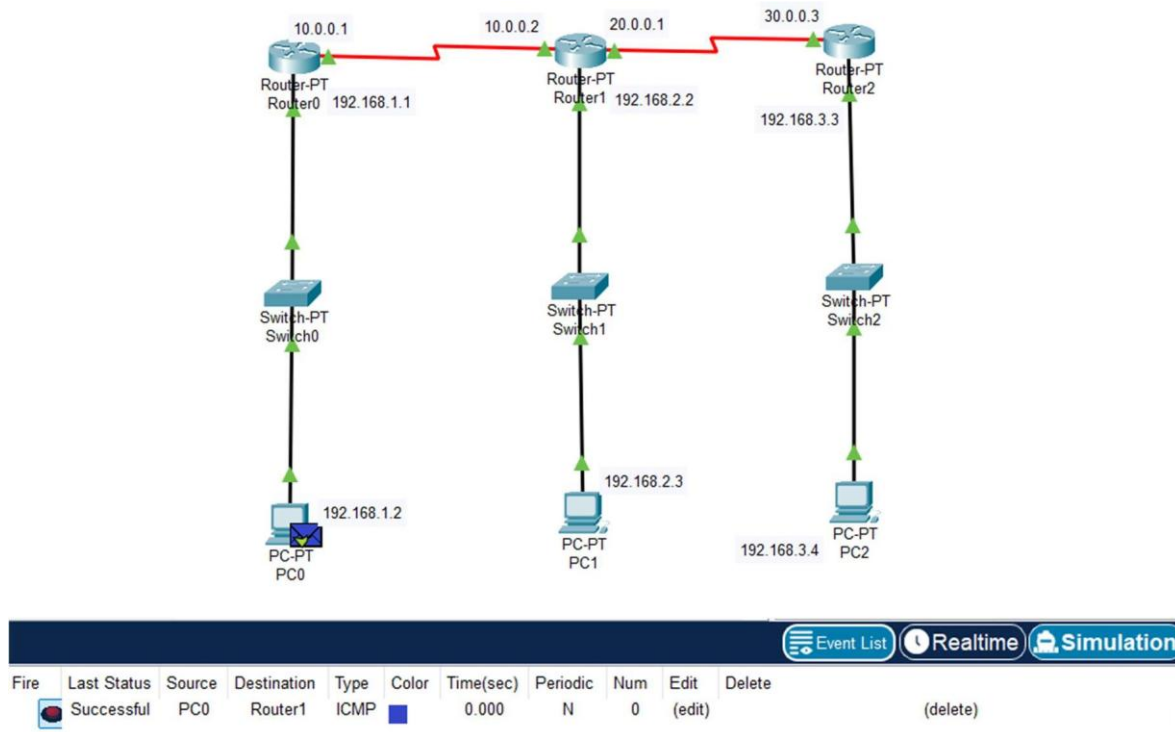
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPsec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoE, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP

Edit Filters Show All/None

Event List Realtime Simulation

Event List		
Vis.	Time(sec)	Last Device
	0.000	--
	0.001	PC0
	0.002	Switch0
	0.003	Router0
	0.004	Router1
	0.005	Router0
	0.006	Switch0
	0.080	--
Visible	0.081	Switch1

6. After it done successfully we see like in shown the picture final step.



7. Verification:

- Open the **Command Prompt** on any PC and use the `ping` command to test communication between networks.
- Use commands like `ping` to verify the communication between devices on different networks.

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.1.1.2

Pinging 10.1.1.2 with 32 bytes of data:

Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time=6ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128
Reply from 10.1.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 10.1.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 6ms, Average = 1ms
```